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Project Title:	Measurement of Flow Field Distortion and Particle Equation of Motion in Turbulent Gas Flows		
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1. The first task is to conduct global measurements of turbulence modification and preferential concentration in the homogeneous and isotropic turbulence chamber. The experiments will be conducted in the laboratory and in micro-gravity to examine the effect of mean particle wakes due to gravitational settling.
2. The second task is to develop a high resolution hybrid PIV/PTV system that can spatially resolve down to the smallest eddy in the flow, and conduct local measurements. The system must be capable of measuring the fluid velocity in the vicinity of particles, where large amounts of light are scattered. It must also be able to estimate the local energy dissipation field around particles.

The present research aims at investigating the physical mechanism of turbulence modification by particles. For this purpose we have developed a turbulence chamber that creates the most basic type of turbulent flow, which is homogeneous and isotropic turbulence without mean flow. Global measurements are first needed to increase the

	understanding of the overall phenomenon. Turbulence modification experiments are needed in the laboratory where gravity exists and in micro-gravity on NASA's KC-135 to evaluate the effects of the mean particle wakes. Preferential concentration of particles also must be examined. Second, local measurements of both phases in the vicinity of individual particles need to be made. A high resolution hybrid PIV/PTV system will enable us to calculate the aerodynamic force on the particles and investigate the particle equation of motion.
Task Description:	1. Gravity plays an important role in fluid-particle interaction in turbulent flows. Eliminating this parameter will give us a better understanding of this complex phenomenon. The turbulence chamber that has been developed is capable of creating turbulence with high Taylor microscale Reynolds numbers and no inflow or outflow. It is ideal for use in a microgravity environment where space is limited. The ideal case of homogeneous and isotropic turbulence will provide a database for modelers and a basis of comparison with ongoing direct numerical simulations (DNS). Such a chamber could also be used for other turbulence research being conducted in a microgravity environment. 2. The high resolution hybrid PIV/PTV system that has been developed enables us to calculate the aerodynamic force on the particles and investigate the particle equation of motion. The results will be used to test various models for turbulence modification in particle-laden flows. We believe it will be able to play a significant role in resolving controversy about the mechanisms and modeling of turbulence attenuation by particles. 3. Energy dissipation is one of the most important factors to clarify the mechanisms of turbulence attenuation. The sink of turbulence energy is generally due to ordinary fluid viscous dissipation and extra dissipation caused by local distortion of the flow field around particles. Previous experiments and simulations have not been capable of resolving this extra dissipation. The present experiment intends to quantify the dissipation caused by local velocity disturbances in the vicinity of the small particles in homogeneous and isotropic flow using high-resolution particle image velocimetry (PIV). The spatial resolution of our custom PIV system is approximately one seventh of the particle diameter and this allows us to estimate the local dissipation field around particles.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Solid particles can modulate the gas-phase statistics in particle-laden flows. Under certain conditions the turbulence is greatly attenuated, even at relatively small mass loadings of particles. Due to the complexity of the fluid-particle interactions, the physical mechanisms of this attenuation are not well understood. None of the commonly used simulation models can correctly predict turbulence modification. One of the critical reasons is that local distortions around particles are not incorporated in most of the models. Obtaining the physics of local distortion around particles is necessary to clarify turbulence modification. The present project examined two fundamental particle-laden flows: homogeneous isotropic turbulence and fully developed turbulent channel flow. The first experiment on particle-laden homogeneous isotropic (HI) turbulence is documented in Hwang and Eaton (2004). A particle image velocimetry (PIV) system was specially developed to be used in multiphase flow environments. Two different types of particles, 165 micron glass beads and 250 micron polystyrene beads, were used because of their similar Stokes numbers yet different particle Reynolds numbers. Measurements have been conducted with two different PIV arrangements. One allowed global measurements of the turbulence modifications. They second system provided much higher spatial resolution showing the local distortion of turbulence around particles. The global measurements were conducted in two different environments, a ground-based environment and under micro-gravity. This allowed study of the effect of gravity, which induces wakes of particles. Experimental results show that gravitational motion of particles adds additional energy to the turbulence and imposes a non-isotropic condition. Turbulence attenuation by particles is greater in micro gravity environments because there is an energy add term by falling particles. A second set of experiments in the HI turbulence chamber examined preferential concentration of particles with Stokes numbers near unity using 25 micron lycopodium and 20 micron and 50 micron glass particles. (Wood et. al., 2005). These data provide quantitative measurement of preferential concentration and direct comparisons to simulation. Local measurements were conducted with high resolution PIV with 70 micron vector spacing, to examine the micro-scale physics of particle-turbulence interactions. The resolution is smaller than the Kolmogorov scale of 100 micron allowing us to obtain accurate measurements of the turbulence kinetic energy (TKE) dissipation. The TKE budget was carefully examined, by introducing a new method to estimate dissipation. The subgrid scale (SGS) stress, which is modeled in Large Eddy Simulation (LES), was directly extracted from the high resolution PIV measurement. Additional LDA measurements of a particle-laden fully-developed downward turbulent channel flow have been conducted (Benson et.al, 2003) to assess the effects of wall roughness on the relative velocity between the gas and particles. The tests were conducted with three different wall roughness conditions all at a channel Reynolds number of 13,800, corresponding to a centerline gas-phase velocity of 10.5 m/s. A dilute loading of 165 micron particles were added to the flow at a mass loading ratio of 15%. Wall roughness had a substantial impact on gas-phase mean velocities across most of the channel width, except very near the wall.
Bibliography Type:	Description: (Last Updated: 08/22/2007)
Articles in Peer-reviewed Journals	Wood AM, Huang W, Eaton JK. " Preferential concentration of particles in homogeneous and isotropic turbulence." International Journal of Multiphase Flow 2005 Oct-Nov;31(10-11):1120-30. , Oct-2005
Articles in Peer-reviewed Journals	Hwang W, Eaton JK. "Homogeneous and isotropic turbulence modulation by small heavy (St~50) particles." Journal of Fluid Mechanics 2006 Oct;564:361-93. http://dx.doi.org/10.1017/S0022112006001431 , Oct-2006
Articles in Peer-reviewed Journals	Hwang W, Eaton JK. "Turbulence attenuation by small particles in the absence of gravity." International Journal of Multiphase Flow, 2006 Dec;32(12):1386-96. http://dx.doi.org/10.1016/j.ijmultiphaseflow.2006.06.008 , Dec-2006

Books/Book Chapters	Eaton JK. "Turbulence modulation by particles." in "Multiphase Flow Handbook." Ed. C. T. Crowe. Boca Raton, FL : CRC Press, 2006. pp 12-86-12-98., Jan-2006